

Ischemic Heart Disease (IHD/ACS) Part 1

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.



Acknowledgements

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Special thanks to the previous scholars.

Most importantly, thank you Dr. Mourad for allowing me to collaborate on this lecture!

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A white speaker icon with sound waves, positioned over the text "Innovate." in the bottom right corner.

Learning Objectives

At the end of this session, students should be able to:

1. Illustrate the role of modifiable and non-modifiable risk factors in atherosclerotic coronary artery disease and key preventive measures
2. Distinguish cardiac from non-cardiac chest pain
3. Differentiate stable angina from unstable angina
4. Assess the need for PCI versus CABG
5. Demonstrate knowledge of medications used in management of angina pectoris



Definitions

- **Angina Pectoris:** Or Angina for short, refers to chest discomfort that occurs when myocardial oxygen demand exceeds oxygen supply.
 - **Stable** Vs. Unstable Vs. **Vasospastic**



What Is Ischemic Heart Disease (IHD) / Acute Coronary Syndrome (ACS) ?

- **Ischemic heart disease (IHD)**: Also referred to as Coronary Artery Disease, is a **chronic** condition associated with an **inadequate supply of blood to the myocardium** due to obstruction of the coronary arteries (usually from atherosclerosis).
- **Acute Coronary Syndrome (ACS)**: An **acute** form of IHD, characterized by a **sudden and severe decrease in blood flow** to the heart, causing symptoms such as **chest pain, shortness of breath, etc.**



What Is Ischemic Heart Disease (IHD) / Acute Coronary Syndrome (ACS) ?

While all ACS are types of IHD, not all IHD is ACS

Ischemic Heart Disease (IHD)

- **Stable Angina**
- Unstable Angina
- **Vasospastic Angina**
- Silent Myocardial Ischemia
- NSTEMI
- STEMI

Acute Coronary Syndrome (ACS)

- Unstable Angina
- NSTEMI
- STEMI



Epidemiology

- Coronary heart disease is the most common type of heart disease. It killed 371,506 people in 2022.
- About **1 in 20 adults** age 20 and older have CAD (about 5%).
- In 2022, about 1 out of every 5 deaths from cardiovascular diseases (CVDs) was among adults younger than 65 years old.



Source:

1. National Center for Health Statistics. Multiple Cause of Death 2018–2022 on CDC WONDER Database. Accessed May 3, 2024. <https://wonder.cdc.gov/mcd.html>
2. Martin SS, Adaj AW, Almarzooq ZI, et al.; American Heart Association Council on Epidemiology and Prevention Statistics Committee; Stroke Statistics Subcommittee. 2024 heart disease and stroke statistics: a report of US and global data from the American Heart Association. *Circulation*. 2024;149:e347–913.
3. Tsao CW, Aday AW, Almarzooq ZI, et al. Heart Disease and Stroke Statistics—2023 Update: A Report From the American Heart Association. *Circulation*. 2023;147:e93–e621.

Risk Factors

MODIFIABLE

- **Cigarette smoking**
- **Hypertension**
- **Hypercholesterolemia**
- **Diabetes Mellitus**
- **Obesity**
- Sedentary Lifestyle/lack of Exercise
- Unhealthy Diet

NON-MODIFIABLE

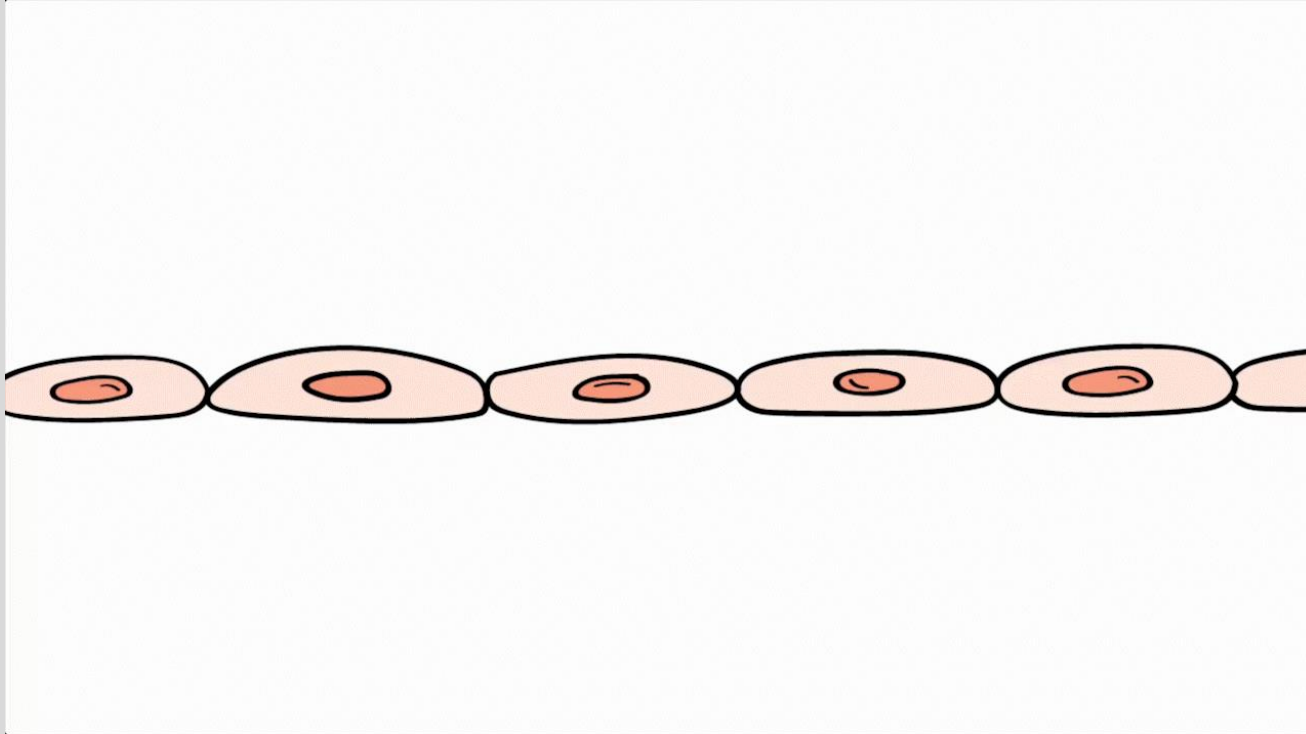
- Age
- Sex
- Ethnicity
- Family History
- Other Medical Conditions



Pathogenesis

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Source:

1. National Center for Health Statistics. Multiple Cause of Death 2018–2022 on CDC WONDER Database. Accessed May 3, 2024. <https://wonder.cdc.gov/mcd.html>
2. Martin SS, Aday AW, Almarazooq ZI, et al.; American Heart Association Council on Epidemiology and Prevention Statistics Committee; Stroke Statistics Subcommittee. 2024 heart disease and stroke statistics: a report of US and global data from the American Heart Association. *Circulation*. 2024;149:e347–913.
3. Tsao CW, Aday AW, Almarazooq ZI, et al. Heart Disease and Stroke Statistics—2023 Update: A Report From the American Heart Association. *Circulation*. 2023;147:e93–e621.
4. Gomez J. (2017). *Other — exact date unavailable*. *Pathogenesis of Atherosclerosis* [Video]. YouTube. <https://www.youtube.com/watch?v=N331s8eFY>

Our Patient:

Chief Complaint (CC): “Chest Pain”

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Defining Chest Pain

2021 AHA/ACC Chest Pain Classification

<u>Category:</u>	<u>Definition:</u>	<u>Clinical Use:</u>
Cardiac Chest Pain (Ischemic)	Pain due to myocardial ischemia (from CAD or coronary vasospasm)	Includes stable angina, unstable angina, NSTEMI/STEMI
Possible Cardiac Chest Pain	Symptoms may represent cardiac ischemia, but are not classic or specific enough	Intermediate-risk cases needing further evaluation
Noncardiac Chest Pain	Clearly not ischemic, or due to other identifiable noncardiac causes	Examples: GERD, costochondritis, panic disorder



Our Patient:

Chief Complaint (CC): “Chest Pain”

History of Present Illness (HPI): A 55-year-old male with a PMHx of HTN and DM type 2 presents to the ED with a 6-month history crushing pain in the middle of his chest. He states that the pain is exacerbated when he goes up the stairs but is relieved with rest. He rates the pain as a 6/10. The pain is present for 7 minutes when he is active. His medications include 20 mg of oral lisinopril daily and 500mg of oral metformin daily. His family history is significant for an MI in his father at the age 50.

What are some differentials for this patient’s presentation?



Differential Diagnosis

PSYCHOGENIC

- Panic Disorder
- Somatic Symptom Disorder
- Illness Anxiety Disorder

PULMONARY

- Pulmonary embolism
- Pneumothorax
- Pneumonia

MUSCULOSKELETAL

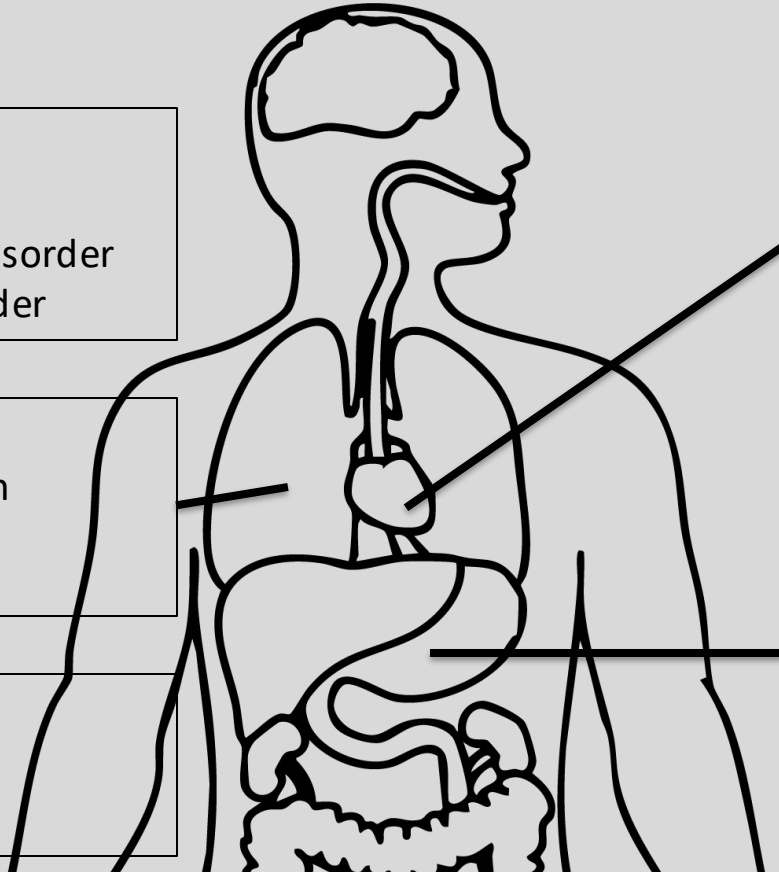
- Costochondritis
- Tietze Syndrome

CARDIAC

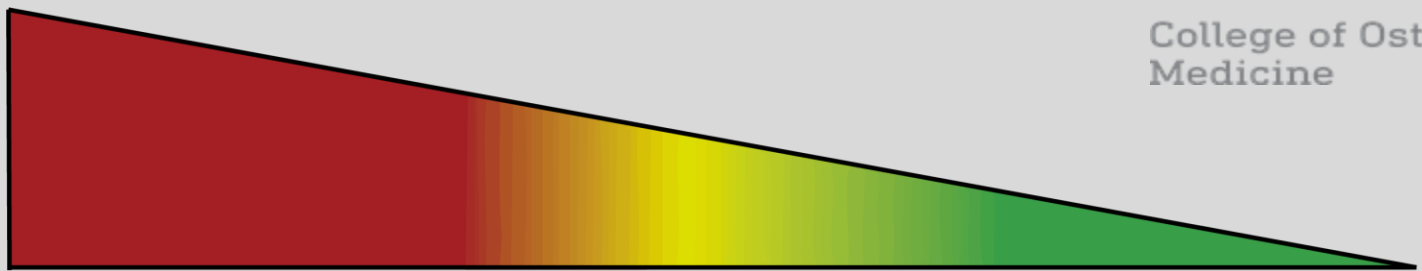
- Stable Angina
- Pericarditis
- Aortic Dissection
- Heart Failure

GASTROINTESTINAL

- Gastroesophageal Reflux Disease (GERD)
- Esophageal Dysphagia
- Gastritis
- Biliary Colic
- Peptic Ulcer Disease (PUD)
- Pancreatitis

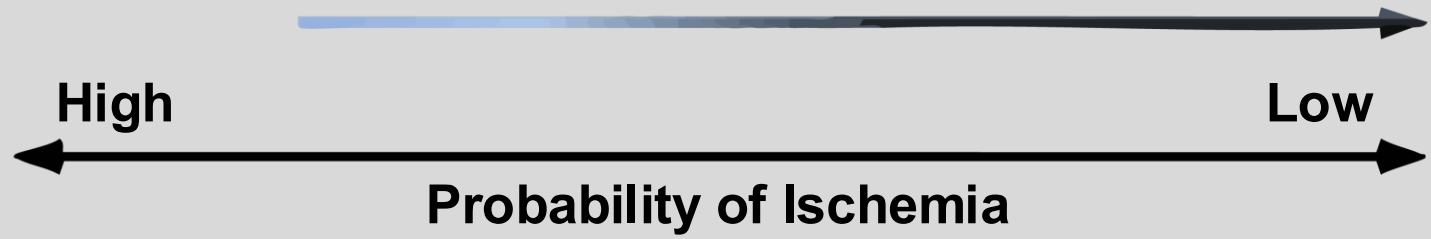


Is Chest Pain Ischemic In Origin?



- | | | | | |
|--|--|--|---|--|
| <ul style="list-style-type: none">• Central• Pressure• Squeezing• Gripping• Heaviness• Tightness• Exertional/Stress Related• Retrosternal | <ul style="list-style-type: none">• Left-sided• Dull• Aching | <ul style="list-style-type: none">• Stabbing | <ul style="list-style-type: none">• Right-sided• Tearing• Ripping• Burning | <ul style="list-style-type: none">• Sharp• Fleeting• Shifting• Pleuritic• Positional |
|--|--|--|---|--|

Consider Likely Non-cardiac Chest Pain



Source:
1. 2021AHA/ACC/AHA/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain J Am Coll Cardiol. 2021 Nov; 78 (22) e187–e285

Women With Chest Pain

- More likely Underdiagnosis/Misdiagnosis
- Differences:
 - Present without chest pain.
 - More likely to have chest pain induced by rest, sleep and mental stress.
 - More often present with accompanying symptoms:
 - Fatigue, Dyspnea, Nausea/Vomiting, Dizziness, Epigastric pain, Jaw/neck pain, Etc.



Source:

1. Gulati M, Cooperman RS, Thorne J, et al. Clinical features and diagnosis of coronary heart disease in women. *UpToDate*. Waltham, MA: UpToDate Inc. <https://www.uptodate.com/contents/clinical-features-and-diagnosis-of-coronary-heart-disease-in-women>. Accessed June 12, 2025.

2. Doylestown Health. (2022, July 28). Chest pain: Could it be a GI condition? Penn Medicine Doylestown Health. Retrieved from Doylestown Health Blog

Our Patient:

Chief Complaint (CC): “Chest Pain”

History of Present Illness (HPI): A 55-year-old male with a PMHx of HTN and DM type 2 presents to the ED with a 6-month history **crushing pain** in the **middle of his chest**. He states that the pain is **exacerbated when he goes up the stairs** but is **relieved with rest**. He rates the pain as a 6/10. The pain is present for 7 minutes when he is active. His medications include 20 mg of oral lisinopril daily and 500mg of oral metformin daily. His family history is significant for an MI in his father at the age 50.

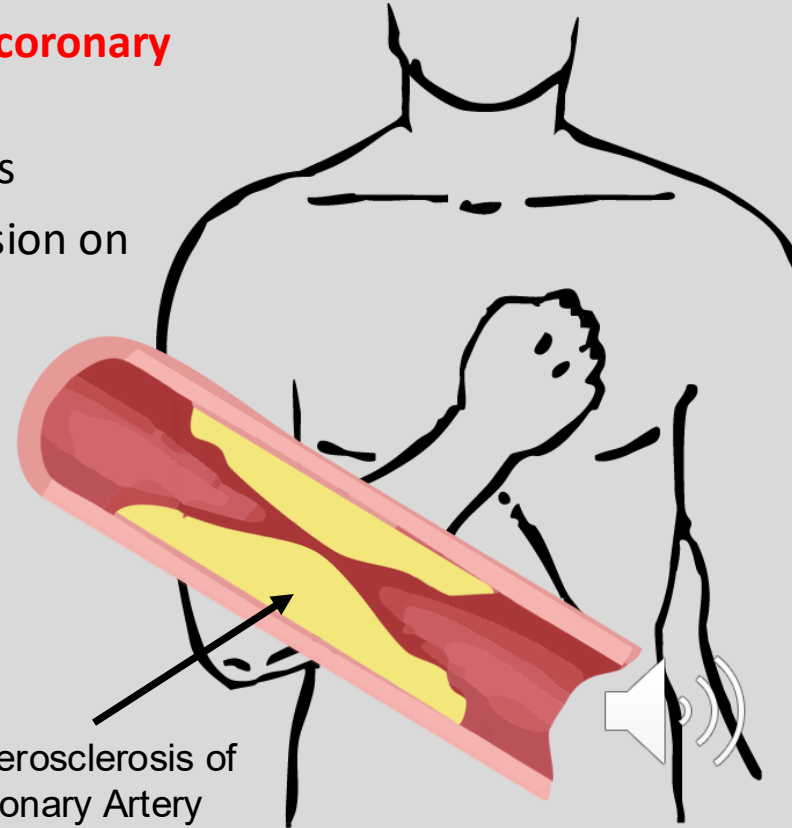


Stable Angina

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- Usually 2° to atherosclerosis (>70% occlusion) of the coronary artery
- Commonly seen in Men >50 years or woman >60 years
- Episodes of chest discomfort (possibly with ST depression on ECG)
 - Heaviness, Pressure, Squeezing
- Levine's sign: Clenched fist placed over the sternum
- Lasts **2-5 minutes, not more than 10**
- Worse with **Exertion** or emotion (e.g stress)
 - Patient has no symptoms without provocation
- Gets better with **Rest** or **sublingual nitroglycerin**



Atherosclerosis of
Coronary Artery

Stable Angina: Pathophysiology

Myocardial Perfusion-demand

Perfusion = Blood Delivered Demand = Blood Required

Oxygen Demand

- Heart rate
- Systolic Blood Pressure
- Myocardial Wall Tension
- Myocardial Contractility

Oxygen Supply

- Coronary Artery Blood Flow
- Hemoglobin Concentration
- Oxygen Unloading



Source:

1. Jameson JL, Faud AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Amsterdam EA, Zidar JP. Approach to the patient with suspected angina pectoris. In: UpToDate, DeSanctis RW (Ed), UpToDate. Accessed June 14, 2025; <https://www.uptodate.com/contents/approach-to-the-patient-with-suspected-angina-pectoris>

Stable Angina: Pathophysiology

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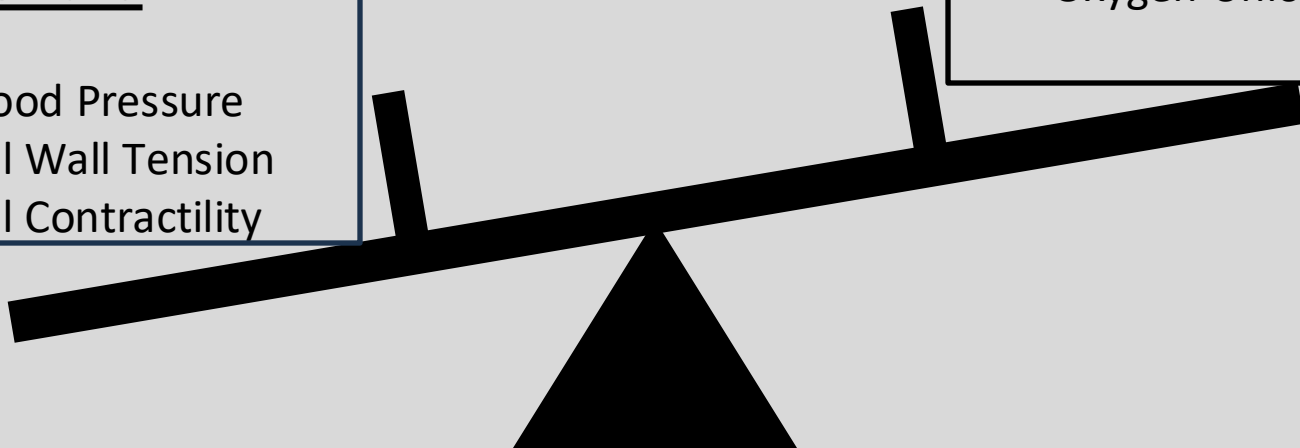
Myocardial Perfusion-demand Mismatch

↓ Oxygen Supply

- ↓ Coronary Artery Blood Flow
- ↔ Hemoglobin Concentration
- ↔ Oxygen Unloading

↑ Oxygen Demand

- ↑ Heart rate
- ↑ Systolic Blood Pressure
- ↑ Myocardial Wall Tension
- ↑ Myocardial Contractility

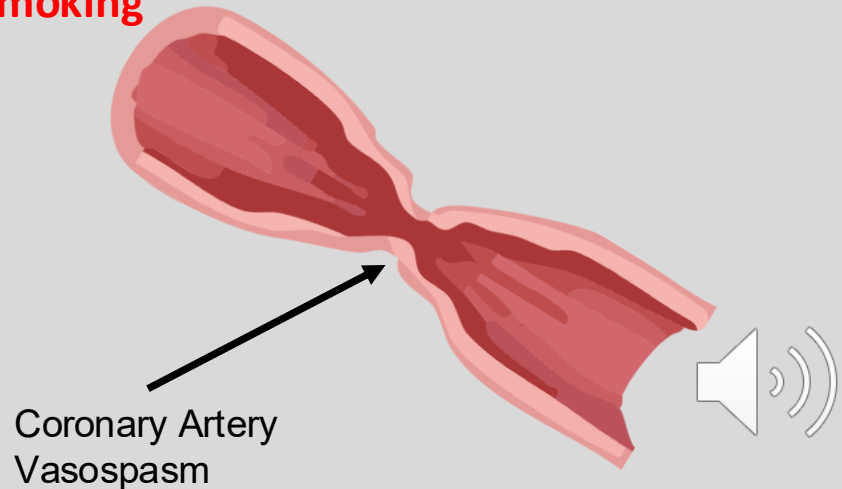


Source:

1. Jameson JL, Faud AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
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Vasospastic Angina (Formally Prinzmetal Angina)

- **2° to coronary artery vasospasm**
- **Transient ST-elevation on EKG**
- Associated with migraines and Raynaud's phenomenon
- **Triggers include cocaine, alcohol and triptans**
- **Relieved by nitroglycerin or calcium channel blockers**
- Treatment: **Calcium Channel Blockers, nitrates, smoking cessation (if applicable)**



Our Patient:

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What is the workup for this patient?



Stable Angina: Workup

- **HISTORY:**

- Location – Diffuse discomfort that may be difficult to localize
- Onset – Gradual onset/offset
- Palliation – Rest
- Provocation – Activity
- Progression –
- Quality – Angina is usually characterized more as a constant discomfort
- Radiation – Shoulders, arms, neck, jaw, ...
- Severity –
- Setting –
- Timing – Lasts 2 to 5 minutes
- Associated Symptoms – Exertional shortness of breath, nausea, diaphoresis, fatigue, ...



Source:

1. Jameson JL, Faud AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
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Stable Angina: Workup

- **HISTORY:**

- Past Medical History – May have a history of cardiovascular disease
- Family History – Family members have premature cardiovascular disease
- Social History – Smoking, Unhealthy diet, sedentary lifestyle



Stable Angina: Workup

- PHYSICAL EXAM:

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Normal



Source:

1. Jameson JL, Faud AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Amsterdam EA, Zidar JP. Approach to the patient with suspected angina pectoris. In: UpToDate, DeSanctis RW (Ed). UpToDate. Accessed June 14, 2025. <https://www.uptodate.com/contents/approach-to-the-patient-with-suspected-angina-pectoris>

Stable Angina: Workup

- **DIAGNOSTIC TESTS:**

- Electrocardiogram (ECG) – **No baseline EKG changes**
- Chest X-ray – Normal or may find enlarged cardiac silhouette
- Cardiac biomarkers – **Troponin and CK-MB will be normal**
- Lipid Panel – May have elevated LDL



Stable Angina: Workup

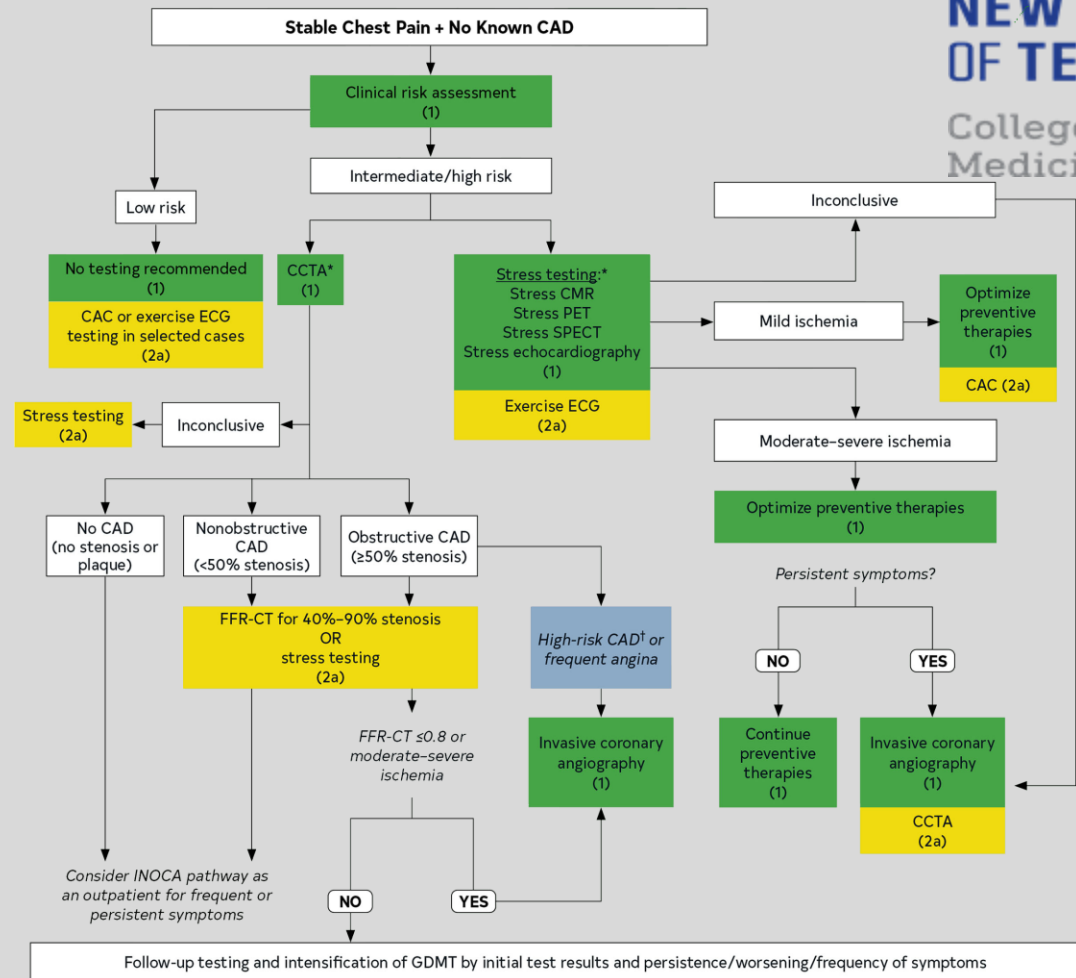
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What else can we do?



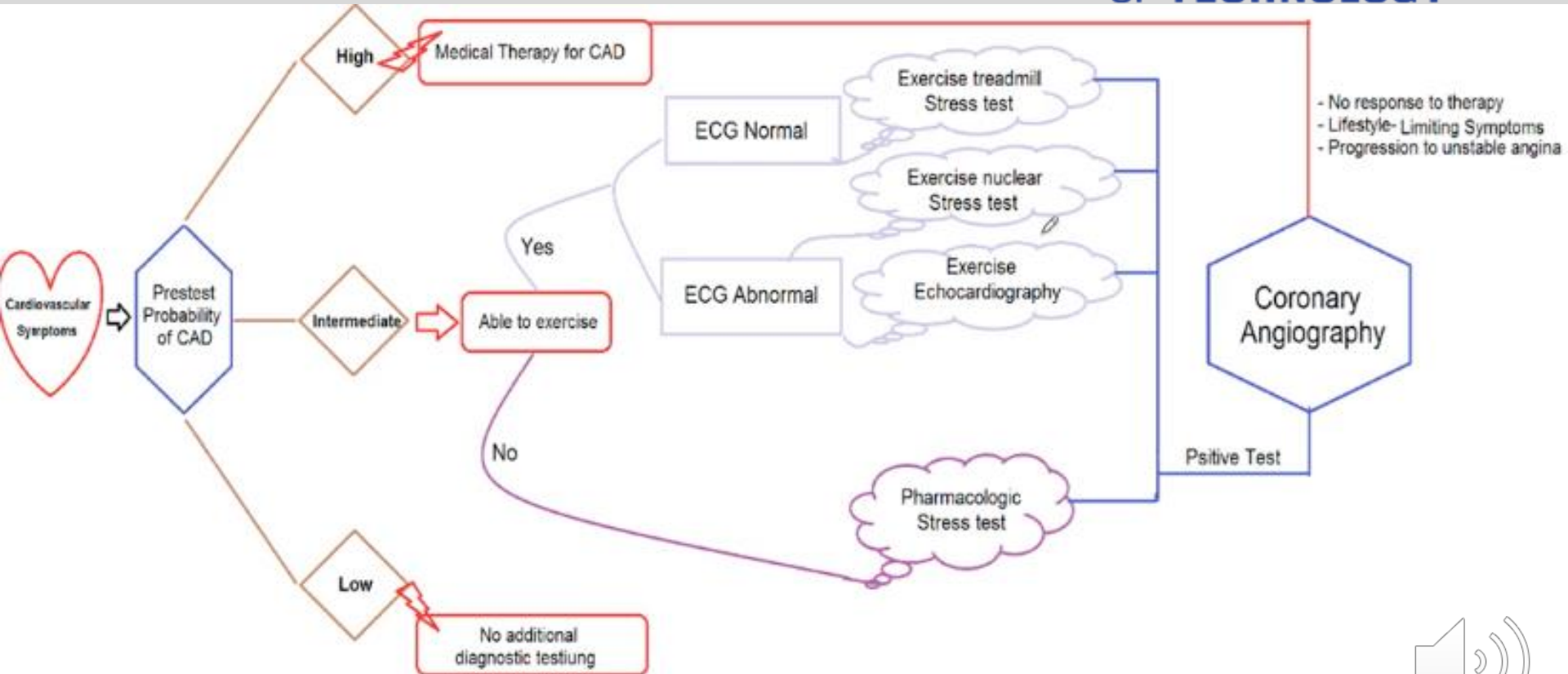
Stress Test



Source: 1. 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain J Am Coll Cardiol. 2021 Nov, 78 (22) e187–e285

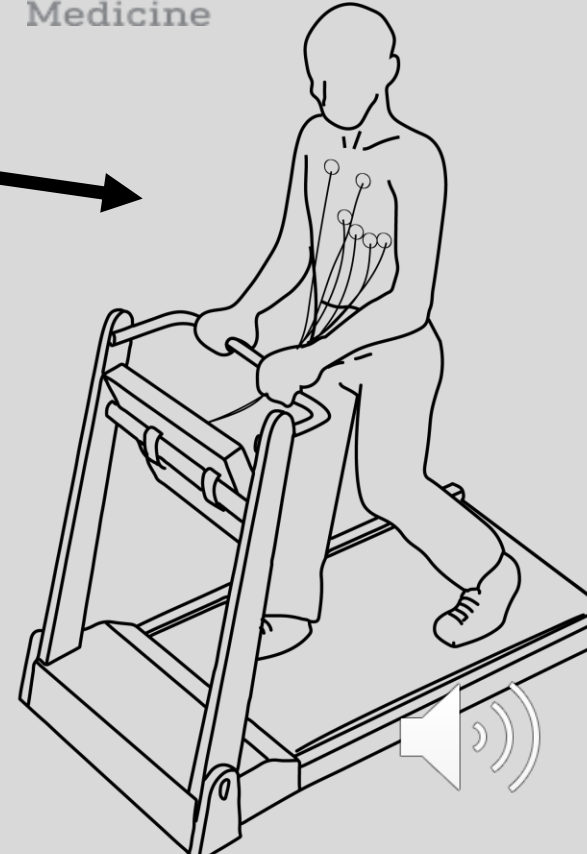


Schematic



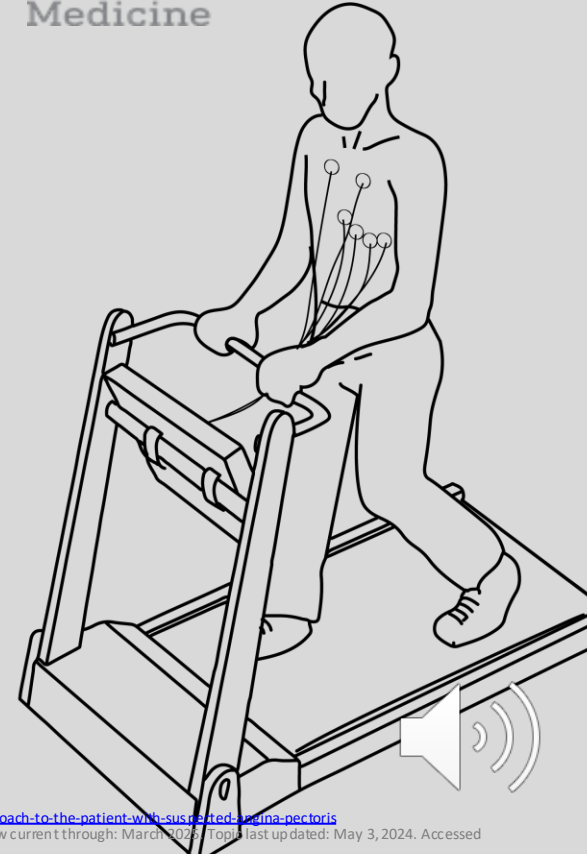
Stress Testing

- Methods of testing
 - Exercise Stress Test
 - Pharmacologic Stress Test
- Evaluation
 - EKG
 - Echocardiogram
 - Radionuclide tracer



Exercise Stress Test

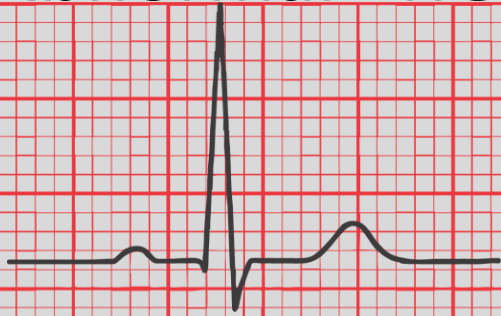
- Patient is asked run/walk on treadmill
- Patient is monitored commonly using an ECG
 - Goal: Recreate and capture transient ischemia
- Positive Stress Test:
 - **ST-segment Depression**
 - **T- wave inversion**
- High Risk Features - warrant discontinuation of test
 - Chest discomfort, severe shortness of breath, dizziness, severe fatigue, Fall in systolic blood pressure >10 mmHg
- Although T-wave abnormalities, conduction disturbances, and ventricular arrhythmias that develop during exercise should be noted, they are also not diagnostic.



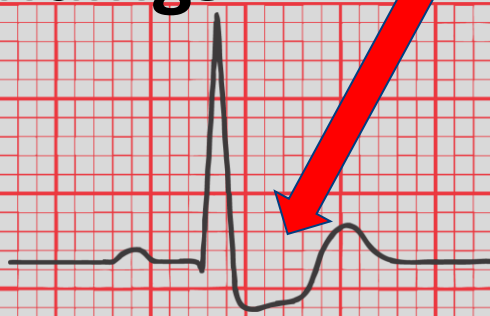
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3. Garber AM, Hlatky MA, Chareonthaitawee P, Askew JW. Stress testing for the diagnosis of obstructive coronary artery disease. In: UpToDate, Kaski JC, Pelikka PA, Dardas TF, eds. Literature review current through: March 2025. Topic last updated: May 3, 2024. Accessed June 15, 2025. <https://www.uptodate.com/contents/stress-testing-for-the-diagnosis-of-obstructive-coronary-artery-disease>

Positive Exercise Stress Test: Abnormal EKG Findings



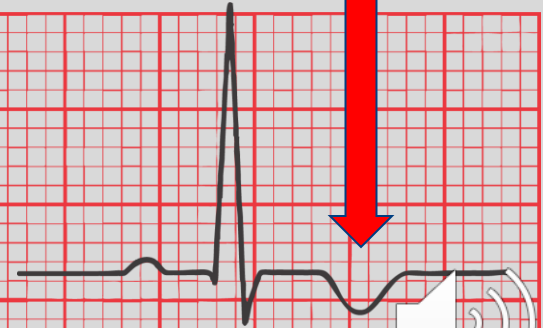
Normal



ST Depression



Normal



T Wave Inversion

Pharmacological Stress Testing

- Performed using **Dobutamine** or vasodilator (**Adenosine**)
 - Adenosine is contraindicated in asthma patients
 - Dobutamine usually paired with echocardiogram
- Patient is monitored using echocardiogram
 - End point: Achievement of target heart rate (85% maximum heart rate)
- Positive Stress Test:
 - Dyskinesia (abnormal ventricular wall movement) on echocardiogram
- High Risk Features - warrant discontinuation of test
 - New/worsening wall motion abnormalities, significant arrhythmias, hypotension (systolic blood pressure > 90 mmHg), severe hypertension, etc



Positive Pharmacological Stress Test: Abnormal Echo Findings

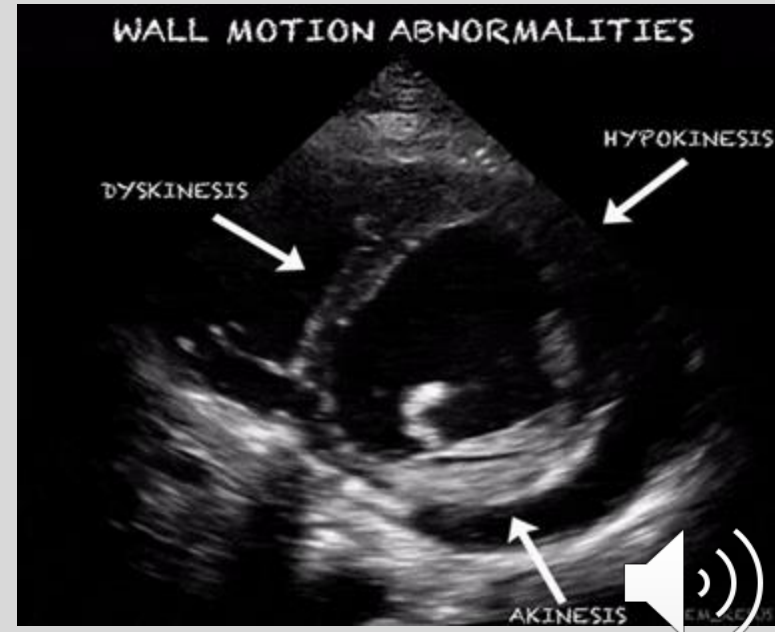
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Normal



Positive Stress Test



Source:

1. 2Garber AM, Hlatky MA, Chareonthaitawee P, Askew JW. Stress testing for the diagnosis of obstructive coronary artery disease. In: UpToDate, Kaski JC, Pelikka PA, Dadas TF, eds. Topic last updated May 3, 2024. Literature review current through March 2025. Accessed June 16, 2025. <https://www.uptodate.com/contents/stress-testing-for-the-diagnosis-of-obstructive-coronary-artery-disease>

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Physical Exam: Within normal limits. Labs: LDL of 200mg/dL (Normal: <100mg/dL), HDL of 40mg/dL (Normal: >50mg/dL). The physician decides to perform an exercise stress test.



Which of the Following EKG Changes Would Denote a Positive Exercise Stress Test?

- A. Peaked T wave
- B. Presence of U wave
- C. PR prolongation
- D. ST segment depression
- E. Widened QRS complex



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- A. Peaked T wave
- B. Presence of U wave
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- D. ST segment depression**
- E. Widened QRS complex



Management

1. Education and Reassurance

2. Identification/Treatment for Aggravating Condition

- Left Ventricular Hypertrophy (LVH), Aortic Valve Disease, Hypertrophic Cardiomyopathy (HCM), Hyperthyroidism, Anemia

3. Lifestyle Modifications

- Diet, Exercise, Smoking Cessation

4. Control for Risk factors

- Hypertension, Diabetes, Dyslipidemia, Obesity

5. Medical Therapy

- **Prevent Angina:** Beta-blockers, Calcium Channel Blockers, Long-acting Nitrates
- **Acute Symptom Relief:** Short-acting Nitrates
- **Prevent Disease Progression:** Antiplatelet therapy, Lipid-lowering therapy, ACE Inhibitors/ARBs, SGLT-2 Inhibitors, GLP-1 Agonist

6. Revascularization

- Percutaneous Coronary Intervention (PCI), Coronary Artery Bypass Graft (CABG)



Treatment to Prevent Angina: Beta-Blockers

- First Line for stable angina
- Nonselective
 - Propranolol
- Cardioselective Drugs (β_1)
 - Atenolol, Metoprolol
- MOA:
 - ↓ Heart Rate (Negative Chronotropy) → ↓ Oxygen Demand
 - ↓ Myocardial Contractility (Negative Inotropy) → ↓ Oxygen Demand
 - ↓ Blood Pressure (↓ Afterload) → ↓ Oxygen demand
 - ↑ Diastolic Filling Time → ↑ Oxygen Supply
- Improve Survival (ONLY with h/o Myocardial Infarction or Systolic Heart Failure)
- DO NOT use with vasospastic angina



Source:

1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Kannam JP, Aroesty JM. Chronic coronary syndrome: Overview of care. In: UpToDate, Lee KL (Ed). Topic last updated [specific update date if known]. Literature review current through [last literature review date]. Accessed June 16, 2025. <https://www.uptodate.com/contents/chronic-coronary-syndrome-overview-of-care>
3. Kannam JP, Gersh BJ. Beta-blockers in the management of chronic coronary syndrome. In: Verheugt F, Dardas TF, eds. UpToDate. Topic last updated May 16, 2023. Literature review current through November 2024. Accessed June 17, 2025. <https://www.uptodate.com/contents/beta-blockers-in-the-management-of-chronic-coronary-syndrome>

Treatment to Prevent Angina: Calcium Channel Blockers

- Used as monotherapy as an alternative to beta-blockers in stable angina
- First line for Vasospastic angina
- Dihydropyridine
 - Nifedipine (Can exacerbate myocardial ischemia due to reflex tachycardia so avoid in patients with unstable angina)
- Non-dihydropyridines
 - Verapamil, Diltiazem
- MOA:
 - ↓ Heart Rate (Negative Chronotropy) → ↓ Oxygen Demand
 - ↓ Myocardial Contractility (Negative Inotropy) → ↓ Oxygen Demand
 - ↑ Coronary Vasodilation → ↑ Oxygen Supply



Source:

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2. Kannam JP, Aroesty JM. Chronic coronary syndrome: Overview of care. In: UpToDate, Lee KL (Ed). Topic last updated [specific update date if known]. Literature review current through [last literature review date]. Accessed June 16, 2025. <https://www.uptodate.com/contents/chronic-coronary-syndrome-overview-of-care>
3. Filippone EJ, Jackson JL. Calcium channel blockers in the management of chronic coronary syndrome. In: UpToDate, Post TW (Ed). UpToDate, Waltham, MA. Updated 2 years ago. Accessed June 17, 2025.

Acute Symptom Management: Short-acting Nitrates

- Sublingual Nitroglycerin
- Onset of Action:
 - Two to five minutes
- Duration:
 - lasts for 15 to 30 minutes
- MOA:
 - ↑ Venous Vasodilation → ↓ Preload (LVEDV) → ↓ Oxygen Demand
- Side Effects:
 - Headache, Flushing, Hypotension
- DO NOT give to patients taking Phosphodiesterase Type 5 (PDE5) Inhibitors (sildenafil, Tadalafil) because of the risk of severe hypotension



Source:

1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Kannam JP, Aroesty JM. Chronic coronary syndrome: Overview of care. In: UpToDate, Lee KL (Ed). Topic last updated [specific update date if known]. Literature review current through [last literature review date]. Accessed June 16, 2025. <https://www.uptodate.com/contents/chronic-coronary-syndrome-overview-of-care>
3. Kannam JP, Gersh BJ. *Nitrates in the management of chronic coronary syndrome*. In: UpToDate, Kaski JC (Ed), UpToDate, Waltham, MA. Literature reviewed through June 2023; last updated September 21, 2022. Accessed June 17, 2025.

Preventing disease Progression: Antiplatelet Therapy

- All patients should be treated with Aspirin
- MOA:
 - COX-1/COX-2 Irreversible inhibitor
- Patients that cannot tolerate Aspirin can be treated with Clopidogrel
- MOA:
 - P2Y₁₂ receptor inhibitor



Source:

1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Kannam JP, Aroesty JM. Chronic coronary syndrome: Overview of care. In: UpToDate, Lee KL (Ed). Topic last updated [specific update date if known]. Literature review current through [last literature review date]. Accessed June 16, 2025. <https://www.uptodate.com/contents/chronic-coronary-syndrome-overview-of-care>
3. Hennekens CH, Sechenova O, Hollar D, Serebruany VL. *Aspirin for the secondary prevention of atherosclerotic cardiovascular disease*. In: UpToDate, Hennekens CH, Cannon CP (Eds), UpToDate, Waltham, MA. Literature reviewed through February 2025; last updated nine months ago. Accessed June 17, 2025.

Preventing disease Progression: Lipid Lowering Therapy (Statins)

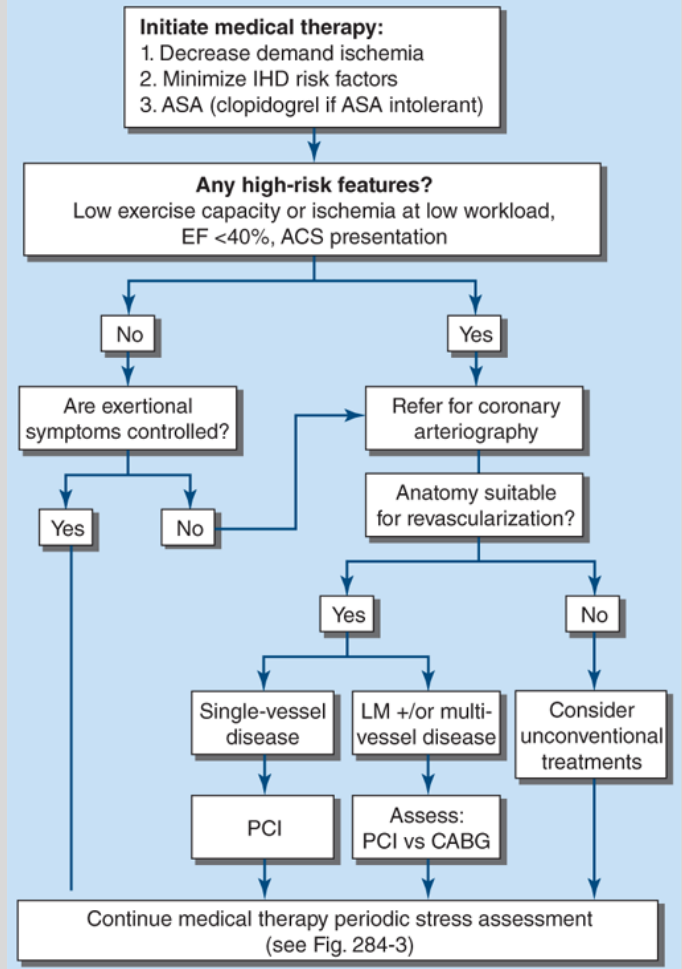
- All Patients should be treated with high-intensity statin
 - Atorvastatin (40-80mg)
 - Rosuvastatin (20-40mg)
- MOA:
 - High-intensity statins reduce LDL-C by $\geq 50\%$
- Check liver enzymes before starting a statin
- Monitor for myopathy



Source:

1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Kannam JP, Aroesty JM. Chronic coronary syndrome: Overview of care. In: UpToDate, Lee KL (Ed). Topic last updated [specific update date if known]. Literature review current through [last literature review date]. Accessed June 16, 2025. <https://www.uptodate.com/contents/chronic-coronary-syndrome-overview-of-care>
3. Martin SS, Blumenthal RS. Management of low-density lipoprotein cholesterol (LDL-C) in the secondary prevention of cardiovascular disease. In: Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. Accessed June 17, 2025. <https://www.uptodate.com/contents/management-of-low-density-lipoprotein-cholesterol-ldl-c-in-the-secondary-prevention-of-cardiovascular-disease>

Revascularization:



Source: Longo D, Fauci A, Kasper D, Hauser S, Jameson JL, Loscalzo J, Holland S, Langford C: Harrison's Principles of Internal Medicine, 22nd Edition
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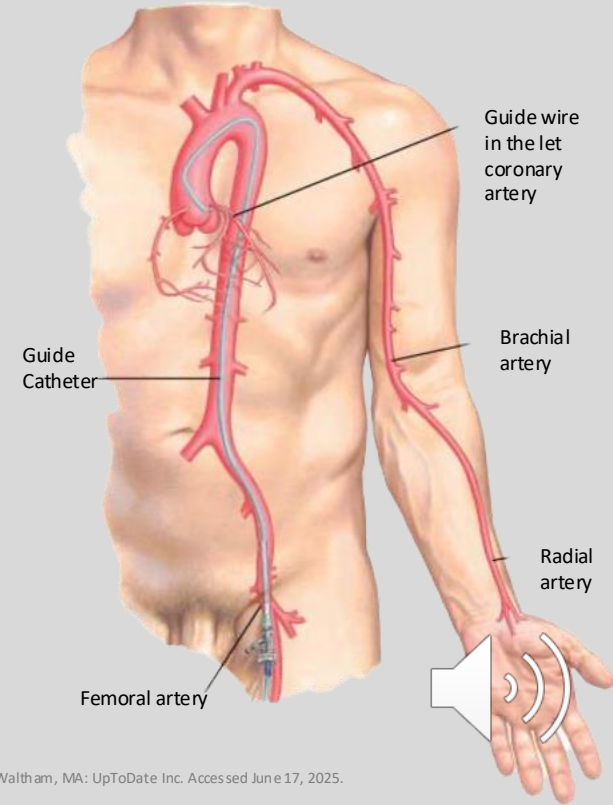
Revascularization:

Percutaneous Coronary Intervention (PCI)

- Used for symptom-limiting angina despite medical therapy
- Performed for **one to two vessel disease**
- Drug-eluting stents (DES) is now standard
- Dual Antiplatelet Therapy (DAPT):
 - Aspirin indefinitely and P2Y12 inhibitor for at least 12 months
- >95% of cases lead to effective angina relief

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Source:

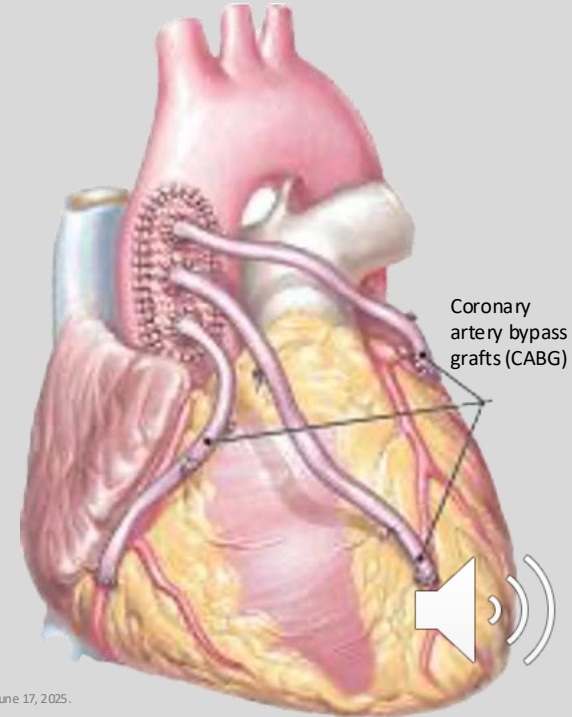
1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Mukherjee D. Revascularization in patients with stable coronary artery disease: Coronary artery bypass graft surgery versus percutaneous coronary intervention. In: Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. Accessed June 17, 2025.
<https://www.uptodate.com/contents/revascularization-in-patients-with-stable-coronary-artery-disease-coronary-artery-bypass-graft-surgery-versus-percutaneous-coronary-intervention>

Revascularization: Coronary Artery Bypass Graft (CABG)

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- Indications:
 - Severe stenoses of **two or three epicardial coronary arteries** with objective evidence of myocardial ischemia
 - Three-vessel disease (with diabetes and/or impaired LV function)
 - Two-vessels including LAD (with diabetes and/or LV disease)
- CABG typically involves:
 - Anastomosis of internal mammary or radial arteries to coronary arteries beyond the obstruction
 - Saphenous vein grafts used if arteries are not viable
- 90% of patients experience abolished or reduced angina after complete revascularization



Source:

1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.
2. Mukherjee D. Revascularization in patients with stable coronary artery disease: Coronary artery bypass graft surgery versus percutaneous coronary intervention. In: Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. Accessed June 17, 2025. <https://www.uptodate.com/contents/revascularization-in-patients-with-stable-coronary-artery-disease-coronary-artery-bypass-graft-surgery-versus-percutaneous-coronary-intervention>

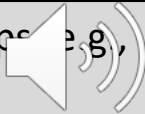
Revascularization: PCI vs CABG

	<u>PCI (Percutaneous Coronary Intervention)</u>	<u>CABG (Coronary Artery Bypass Graft)</u>
Indications	Single or two vessel disease, normal LV function, lifestyle-limiting angina despite optimal medical therapy	Two or three-vessel disease, diabetes with LAD involvement, or impaired LV function
Angina relief	Angina may recur earlier	More durable symptom relief, especially in complex CAD
Restenosis risk	Higher risk from in-stent restenosis or new lesions (reduced with DES)	Lower, especially with arterial grafts
Stroke risk	Lower	Higher
Repeat revascularization	More likely	Less likely
Survival benefit	Less established benefit; varies by complexity	Improves survival in high-risk groups (e.g., left main, diabetics)

Source:

1. Jameson JL, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J, eds. *Harrison's Principles of Internal Medicine*. 22nd ed. McGraw Hill; 2022.

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Question

A 50-year-old female with PMHx of hyperlipidemia presents to the clinic with chest pain for the past 3 months. She describes the pain as “an elephant sitting on my chest.” The pain is brought on by doing chores around the house and relieved with rest. She states the pain does not radiate. The patient states that she smokes ½ pack a day for the past 20 years. Her only medication is 40mg of oral Atorvastatin daily. Vitals and physical exam is normal. There are no changes found on resting ECG. However, on stress testing the patient develops ST segment depression in Leads I and aVL at 85% of her maximal HR. What would be the best appropriate next step in management?

- A. Coronary Artery Bypass Graft (CABG)
- B. Percutaneous Coronary Intervention (PCI)
- C. Start patient on aspirin
- D. Increase the dose of Atorvastatin
- E. Counsel on smoking cessation



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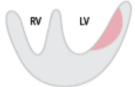
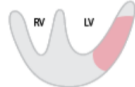
Summary

	<u>Stable Angina</u>	<u>Vasospastic Angina</u>
Pain	Crushing, Squeezing Substernal Pain	Spontaneous crushing chest pain
Exacerbated	Exacerbated with exertion	Unrelated to exertion, can be triggered by cocaine
Relief	Relieved by Rest or Nitroglycerin	Nitroglycerin or CCB
EKG	No EKG findings at rest	Transient ST elevation
Cardiac Biomarkers	None	None
Treatment	PCI, CABG, Lifestyle modification, Nitrates, Beta-blockers/CCB, ACE-Is	CCB are first line



Summary

Ischemic heart disease manifestations

Angina	Chest pain due to ischemic myocardium 2° to coronary artery narrowing or spasm; no necrosis.			
	▪ Stable—usually 2° to atherosclerosis ($\geq 70\%$ occlusion); exertional chest pain in classic distribution (possibly with ST depression on ECG), resolving with rest or nitroglycerin. ▪ Vasospastic (formerly Prinzmetal or variant)—occurs at rest 2° to coronary artery spasm; transient ST elevation on ECG. Tobacco smoking is a risk factor; hypertension and hypercholesterolemia are not. Triggers include cocaine, alcohol, and triptans. Treat with Ca^{2+} channel blockers, nitrates, and smoking cessation (if applicable). ▪ Unstable—thrombosis with incomplete coronary artery occlusion; +/- ST depression and/or T-wave inversion on ECG but no cardiac biomarker elevation (unlike non-ST-segment elevation MI [NSTEMI]); $\uparrow$ in frequency or intensity of chest pain or any chest pain at rest.			
	Stable angina	Unstable angina	NSTEMI	STEMI
	PAIN	On exertion	Mild exertion or at rest	At rest
	TROPONIN LEVEL	No elevation	No elevation	Elevated
INFARCTION	None	None		
			Subendocardial 	Transmural 
ECG CHANGES	None	Possible ST depression and/or T-wave inversion	ST depression and/or T-wave inversion	ST elevation





Questions?

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